**P2.18**

let $f(x) = A e^{ikx} + B e^{-ikx} = C \cos kx + D \sin kx$

Then $A(\cos kx + i \sin kx) + B(\cos kx - i \sin kx) =$

$$= (A+B)\cos kx + i(A-B)\sin kx = C\cos kx + D\sin kx$$

$$\Rightarrow \begin{cases} A+B = C \\ i(A-B) = D \end{cases} \Rightarrow A - B = -iD$$

$$\Rightarrow \begin{cases} A = \dfrac{C-iD}{2} \\ B = \dfrac{C+iD}{2} \end{cases}$$

**P2.19**

The probability current $J = \dfrac{i\hbar}{2m}\left(\psi \dfrac{\partial \psi^*}{\partial x} - \psi^* \dfrac{\partial \psi}{\partial x}\right)$

For $\psi_k(x,t) = A e^{i(kx - \frac{\hbar k^2}{2m}t)}$

$$\psi \frac{\partial}{\partial x}\psi^* = |A|^2 e^{ikx} \frac{\partial}{\partial x} e^{-ikx} = -ik|A|^2$$

$$\psi^* \frac{\partial}{\partial x}\psi = |A|^2 e^{-ikx} \frac{\partial}{\partial x} e^{ikx} = ik|A|^2$$

$$\Rightarrow J(x,t) = \frac{i\hbar}{2m}\left(-ik|A|^2 - ik|A|^2\right) =$$

$$= \frac{\hbar k}{m}|A|^2$$

Note that $J$ is in the direction of momentum $\underline{p = \hbar k}$

Since classically $\dfrac{\hbar k}{m} = \dfrac{p}{m} = v$ (velocity),

$J(x,t)$ does describe the flow of probability.

**P2.21** $\Psi(x,0) = A\, e^{-a|x|}$

(a) $$1 = A^2 \int_{-\infty}^{\infty} dx\, e^{-2a|x|} = 2A^2 \int_0^{\infty} dx\, e^{-2ax} =$$

$$= \frac{2A^2}{2a} \Rightarrow A = \sqrt{a}$$

(b) $$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, \Psi(x,0)\, e^{-ikx} = \sqrt{\frac{a}{2\pi}} \int_{-\infty}^{\infty} e^{-a|x| - ikx}\, dx$$

$$= \sqrt{\frac{a}{2\pi}} \left\{ \int_0^{\infty} e^{-ax-ikx}\, dx + \int_{-\infty}^{0} e^{+ax-ikx}\, dx \right\}$$

$$= \sqrt{\frac{a}{2\pi}} \int_0^{\infty} \left( e^{-ax-ikx} + e^{-ax+ikx} \right) dx \quad \xrightarrow{x' = -x} \quad \int_0^{\infty} e^{-ax'+ikx'}\, dx'$$

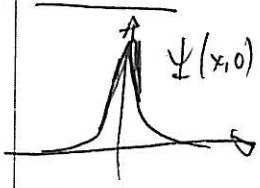
$$= \sqrt{\frac{a}{2\pi}} \left( \frac{1}{a+ik} + \frac{1}{a-ik} \right)$$

$$= \sqrt{\frac{a}{2\pi}}\, \frac{2a}{a^2+k^2}$$

(c) $$\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{\frac{a}{2\pi}}\, \frac{2a}{a^2+k^2}\, e^{i\left(kx - \frac{\hbar k^2}{2m} t\right)}\, dk$$

$$= \frac{a^{3/2}}{\pi} \int_{-\infty}^{\infty} \frac{e^{i\left(kx - \frac{\hbar k^2}{2m} t\right)}}{a^2+k^2}\, dk$$

(d) $\underline{a \to \infty}$ $\Rightarrow$ $\Psi(x,0) = \sqrt{a}\, e^{-a|x|}$ is very <u>sharply</u> peaked near $x=0$.




but $\phi(k) = \sqrt{\frac{2}{\pi}}\, \frac{a^{3/2}}{a^2+k^2} \to \sqrt{\frac{2}{\pi}}\, \frac{1}{\sqrt{a}}$ for $a \to \infty$ almost does not depend on $\underline{k}$ $\Rightarrow$ flat function!

$\underline{a \to 0}$ $\Rightarrow$ $\Psi(x,0) \cong \sqrt{a}$ (almost constant) but $\phi(k) \approx \sqrt{\frac{2}{\pi}}\, \frac{a^{3/2}}{k^2}$ is sharply peaked near $k=0$.

P2.22 $\Psi(x,0) = A e^{-ax^2}$

(a) $$1 = A^2 \int_{-\infty}^{\infty} e^{-2ax^2}\,dx = A^2\sqrt{\frac{\pi}{2a}} \Rightarrow A = \left(\frac{2a}{\pi}\right)^{1/4}$$

(b) Find $\phi(k)$ and then $\Psi(x,t)$:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \left(\frac{2a}{\pi}\right)^{1/4} e^{-ax^2} e^{-ikx} =$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{2a}{\pi}\right)^{1/4} \int_{-\infty}^{\infty} dx\, e^{-a\left(x+\frac{ik}{2a}\right)^2 - \frac{k^2}{4a}}$$

Complete the square $ax^2 + ikx = a\left(x^2 + \frac{ik}{a}x\right) = a\left[\left(x+\frac{ik}{2a}\right)^2 - \left(\frac{ik}{2a}\right)^2\right] =$

$$= a\left(x+\frac{ik}{2a}\right)^2 + \frac{k^2}{4a}$$

Shift the integration variable $x + \frac{ik}{2a} = y$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \left(\frac{2a}{\pi}\right)^{1/4} \underbrace{\int_{-\infty}^{\infty} dy\, e^{-ay^2}}_{\sqrt{\frac{\pi}{a}}} e^{-\frac{k^2}{4a}} =$$

$$= \frac{1}{\sqrt{2a}} \left(\frac{2a}{\pi}\right)^{1/4} e^{-\frac{k^2}{4a}} = \frac{1}{(2\pi a)^{1/4}} e^{-k^2/4a}.$$

Important point: Fourier transform of Gaussian $e^{-ax^2}$ is Gaussian in momentum space, $e^{-k^2/4a}$!

Next $\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k)\, e^{i\left(kx - \frac{\hbar k^2}{2m}t\right)}\, dk$

$$= \frac{1}{\sqrt{2\pi}} \frac{1}{(2\pi a)^{1/4}} \int_{-\infty}^{\infty} dk\, e^{-\frac{k^2}{4a}}\, e^{i\left(kx - \frac{\hbar k^2}{2m}t\right)}\, dk$$

The argument of exponential reads:

$$\frac{k^2}{4a} - ikx + \frac{i\hbar t k^2}{2m} = \frac{k^2}{4a}\left(1 + i\frac{2a\hbar t}{m}\right) - ikx =$$

$$= \frac{(1 + 2ia\hbar t/m)}{4a}\left(k^2 - i\,x\,\frac{4a}{1+i2a\hbar t/m}\,k\right) =$$

$$= \frac{(1+2ia\hbar t/m)}{4a}\left\{\left(k - i\,\frac{2xa}{1+i2a\hbar t/m}\right)^2 - \left(\frac{i\,2xa}{1+i2a\hbar t/m}\right)^2\right\}$$

$$= \frac{1+2ia\hbar t/m}{4a}\left(k - i\,\frac{2xa}{1+i2a\hbar t/m}\right)^2 + \frac{x^2 a}{1+i2a\hbar t/m}$$

Thus $\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \frac{1}{(2\pi a)^{1/4}}\, e^{-\frac{ax^2}{1+i2a\hbar t/m}} \underbrace{\int_{-\infty}^{+\infty} dp\, e^{-p^2\frac{1+2ia\hbar t/m}{4a}}}_{\sqrt{\frac{4a\cdot\pi}{1+i2a\hbar t/m}}}$

where $p = k - i\,\frac{2xa}{1+i2a\hbar t/m}$

$$\Psi(x,t) = \left(\frac{2a}{\pi}\right)^{1/4} \frac{1}{\sqrt{1+2ia\hbar t/m}}\, e^{-\frac{ax^2}{1+2ia\hbar t/m}}$$

(c) To find $|\Psi(x,t)|^2$ we first denote $y = \frac{2a\hbar t}{m}$

Then $\Psi(x,t) = \left(\frac{2a}{\pi}\right)^{1/4} \frac{1}{\sqrt{1+iy}} e^{-\frac{ax^2}{1+iy}}$

$$\Psi^*(x,t) = \left(\frac{2a}{\pi}\right)^{1/4} \frac{1}{\sqrt{1-iy}} e^{-\frac{ax^2}{1-iy}}$$

Thus $|\Psi(x,t)|^2 = \left(\frac{2a}{\pi}\right)^{1/2} \frac{1}{\sqrt{1+y^2}} e^{-ax^2\left(\frac{1}{1+iy} + \frac{1}{1-iy}\right)} =$

$$= \left(\frac{2a}{\pi}\right)^{1/2} \frac{1}{\sqrt{1+y^2}} e^{-\frac{2ax^2}{1+y^2}}$$

In terms of $w = \sqrt{\frac{a}{1+(2\hbar a t/m)^2}}$ we have $w = \sqrt{\frac{a}{1+y^2}}$

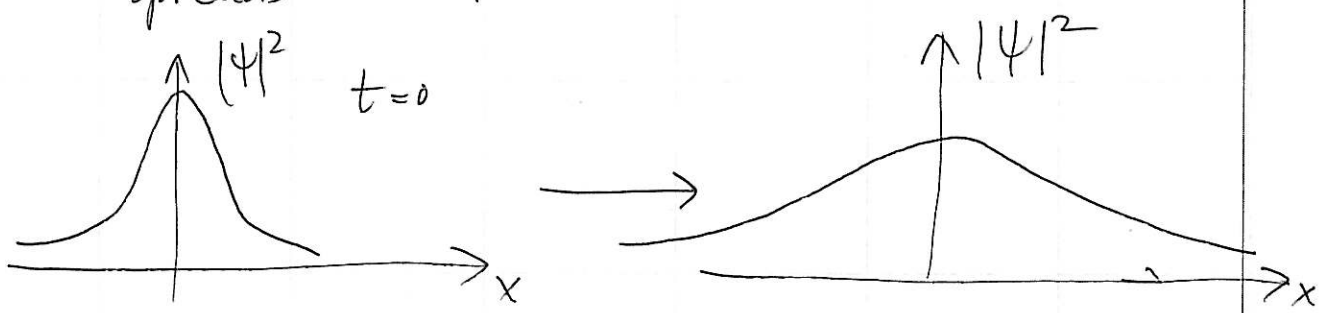
$$|\Psi(x,t)|^2 = \left(\frac{2}{\pi}\right)^{1/2} w\, e^{-2w^2x^2}$$

Note that $\frac{1}{w}$ is the half-width of $x$

and since $w \xrightarrow[t\to\infty]{} 0$, $\frac{1}{w} \xrightarrow[t\to\infty]{} \infty$, which

means that initially sharp $|\Psi(x,0)|^2$ $\left[\text{half-width } \frac{1}{\sqrt{a}}\right]$

spreads out:




But the area under $|\Psi(x,t)|^2$ curve is always 1!

(d) we found $|\Psi(x,t)|^2$ to be <u>even</u> function of $x$ (Gaussian)

$$\Rightarrow \langle x \rangle = \int_{-\infty}^{\infty} x\,|\Psi(x,t)|^2\,dx = 0 \quad \text{by oddness.}$$

$$\Rightarrow \langle p \rangle = m\frac{d\langle x\rangle}{dt} = 0 \quad \text{as well.}$$

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} x^2\,|\psi(x,t)|^2\,dx = \left(\frac{2}{\pi}\right)^{1/2} w \int_{-\infty}^{\infty} x^2 e^{-2w^2x^2}\,dx$$

But $\displaystyle\int_{-\infty}^{+\infty} x^2 e^{-\alpha x^2}\,dx = -\frac{d}{d\alpha}\int e^{-\alpha x^2}\,dx = -\frac{d}{d\alpha}\sqrt{\frac{\pi}{\alpha}} = \frac{\sqrt{\pi}}{2\alpha^{3/2}}$

$$\Rightarrow \langle x^2 \rangle = \left(\frac{2}{\pi}\right)^{1/2} w\,\frac{\sqrt{\pi}}{2\cdot(2w^2)^{3/2}} = \frac{w\cdot 2^{1/2}}{2^{5/2}\,w^3} = \frac{1}{4w^2} =$$

$$= \frac{1+(2\hbar a t/m)^2}{4a}.$$

$$\langle p^2 \rangle = \int \psi^*(x,t)\left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)^2 \psi(x,t)\,dx =$$

$$= -\hbar^2\left(\frac{2a}{\pi}\right)^{1/2}\frac{1}{\sqrt{1+i\gamma}}\,\frac{1}{\sqrt{1-i\gamma}}\int e^{-\frac{ax^2}{1-i\gamma}}\underbrace{\frac{d^2}{dx^2}e^{-\frac{ax^2}{1+i\gamma}}}_{\frac{d}{dx}\left(-\frac{2ax}{1+i\gamma}e^{-\frac{ax^2}{1+i\gamma}}\right)}\,dx$$

$$= \left(\frac{2a}{\pi}\right)^{1/2}\frac{-\hbar^2}{\sqrt{1+\gamma^2}}\int e^{-\frac{ax^2}{1-i\gamma}}\left[-\frac{2a}{1+i\gamma}+\left(\frac{2ax}{1+i\gamma}\right)^2\right]e^{-\frac{ax^2}{1+i\gamma}}\,dx =$$

$$= \left(\frac{2a}{\pi}\right)^{1/2}\frac{-\hbar^2}{\sqrt{1+\gamma^2}}\int\left(-\frac{2a}{1+i\gamma}+\frac{4a^2x^2}{(1+i\gamma)^2}\right)e^{-\frac{2ax^2}{1+\gamma^2}}\,dx$$

$$= \left(\frac{2a}{\pi}\right)^{1/2}\frac{-\hbar^2}{\sqrt{1+\gamma^2}}\left(-\frac{2a}{1+i\gamma}\sqrt{\frac{\pi(1+\gamma^2)}{2a}}+\frac{4a^2}{(1+i\gamma)^2}\,\frac{\sqrt{\pi}}{2}\left(\frac{1+\gamma^2}{2a}\right)^{3/2}\right)$$

$$\langle p^2 \rangle = \left(\frac{2a}{\pi}\right)^{1/2} (-\hbar^2) \sqrt{\pi} \left( -\frac{\sqrt{2a}}{1+i\gamma} + \frac{2a^2(1+\gamma^2)}{(1+i\gamma)^2 (2a)^{3/2}} \right)$$

$$= (-\hbar^2\sqrt{\pi}) \frac{2a}{\sqrt{\pi}} \left( -\frac{1}{1+i\gamma} + \frac{1+\gamma^2}{(1+i\gamma)^2 2} \right) = -\hbar^2(2a) \cdot$$

$$\cdot \frac{1}{1+i\gamma} \left( -1 + \frac{(1+i\gamma)(1-i\gamma)}{(1+i\gamma)2} \right) = (-2a\hbar^2) \left( \frac{1-i\gamma}{2} - 1 \right) \frac{1}{1+i\gamma} =$$

$$= (-2a\hbar^2) \frac{-1-i\gamma}{2} \frac{1}{1+i\gamma} = (-2a\hbar^2)\left(-\frac{1}{2}\right) = a\hbar^2.$$

Thus $\langle x^2 \rangle = \frac{1}{4w^2}$, $\langle p^2 \rangle = a\hbar^2$

(e) $$\sigma_x = \frac{1}{2w} = \sqrt{\frac{1+(2\hbar a t/m)^2}{4a}}$$

$$\sigma_p = \sqrt{a}\,\hbar$$

$$\Rightarrow \sigma_x \sigma_p = \sqrt{\frac{1+(2\hbar a t/m)^2}{4a}} \cdot \sqrt{a}\,\hbar = \frac{\hbar}{2}\sqrt{1+\left(\frac{2\hbar t a}{m}\right)^2}$$

$\Rightarrow$ Uncertainty is minimal at $t=0$, when $\sigma_x \sigma_p = \frac{\hbar}{2}$ and then grows with time (because $\langle x^2 \rangle$ grows with time!).

At long times

$$\sigma_x \sigma_p \underset{t\to\infty}{\cong} \frac{\hbar}{2} \cdot \frac{2\hbar t a}{m}.$$